



An interactive quiz platform where learners practice, contribute, and help educators continuously improve learning content.

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1. Project Overview

TAMAKAN is a role-based web application designed around a collaborative quiz workflow. It supports two user types—learners and educators—and combines quiz delivery, automatic scoring, ratings and written feedback, learner-generated question recommendations, and educator review. The defining idea is that learners do not only consume assessment content; they can also propose new questions that educators review before accepted recommendations are added to the quiz question bank.

2. Project Objectives

- Provide a structured environment for topic-based quizzes.
- Give learners immediate, percentage-based results after quiz submission.
- Use motivational media to make result feedback more engaging.
- Allow learners to rate quizzes and provide written comments.
- Enable learners to recommend complete multiple-choice questions to educators.
- Keep educators in control of content through approval, rejection, commenting, and question management.
- Use role-based sessions to separate learner and educator experiences.

3. User Roles and Functional Scope

3.1 Learner

- Register and sign in as a learner.
- Browse quizzes and filter them by topic.
- View the educator associated with each quiz and the question count.
- Start and complete multiple-choice quizzes.
- Receive a calculated percentage score.
- Receive score-dependent motivational feedback.
- Rate a completed quiz from 1 to 5 and submit an optional comment.
- Recommend a new question by choosing a topic and educator, entering the question, four answers, the correct answer, and an optional image.
- View previously submitted recommendations, their status, and educator comments.

3.2 Educator

- Register and sign in as an educator.
- Select at least one topic during educator registration.
- Access quizzes associated with educator topics.
- Add, edit, and delete quiz questions.
- Attach an optional figure to a question.
- View aggregate quiz statistics including number of attempts, average score, feedback count, and average rating.
- Read learner feedback.
- Review learner-recommended questions.
- Approve or disapprove a recommendation and add a review comment.
- Automatically copy an approved recommendation into the quiz question table.

4. Core Workflow

The main learning cycle implemented by TAMAKAN is:

Step	Actor	System Behavior
1	Educator	Creates and maintains quiz questions.
2	Learner	Selects a topic/quiz and answers its questions.
3	System	Calculates the percentage score and stores the attempt.
4	System	Displays motivational media according to the score range.
5	Learner	Submits a rating and optional written feedback.
6	Learner	May recommend a new multiple-choice question.
7	Educator	Reviews the recommendation and approves or disapproves it.
8	System	If approved, copies the recommendation into the quiz question bank.

5. Functional Modules

Module	Purpose	Main Files
Authentication	Registration, login, password verification/hashing, role selection, session initialization.	pages/auth/Auth.php, logout.php
Learner Dashboard	Quiz discovery, topic filtering, educator display, recommendation status.	pages/learner/learnerHome.php, api/get_quizzes.php
Quiz Engine	Loads questions, accepts answers, calculates results.	pages/learner/takequiz.php, score.php
Feedback	Stores rating and written feedback after quiz completion.	actions/learner/submit_feedback.php
Question Recommendation	Collects learner-proposed questions and optional images.	pages/learner/recommend_question.php, actions/learner/add_recommendation.php
Educator Dashboard	Question management, aggregate statistics, recommendation review.	pages/educator/educatorHome.php
Question CRUD	Add, edit, and delete educator-managed questions.	pages/educator/AddQ.php, editQ.php; actions/educator/*
Dynamic Data APIs	Loads filtered quizzes and educators without rebuilding the page manually.	api/get_quizzes.php, get_educators_by_topic.php

6. Scoring and Feedback Logic

For a submitted quiz, TAMAKAN compares learner answers with the stored correct answers, counts correct responses, and calculates the percentage as $(\text{correct answers} \div \text{total questions}) \times 100$. The result is stored in the takenquiz table.

Score Range	Feedback Media
80% and above	Applause.mp4
50% to below 80%	keep-going.mp4
Below 50%	Sorry.mp4

The result page also provides a 1–5 rating control and a written feedback field. This feedback is stored separately from the quiz attempt.

7. Learner Recommendation and Educator Review

A learner recommendation is a complete candidate quiz question. The recommendation form records the selected topic/educator through the target quiz, question text, optional figure, four answer choices (A–D), and the correct answer. New recommendations begin with the status 'pending'.

Review Decision	Stored Status	Result
Approve	approved	The recommendation is copied into quizquestion.
Disapprove	disapproved	The recommendation remains recorded but is not copied into quizquestion.

The educator may also save a comment with the decision. The learner dashboard retrieves the recommendation status and educator comment, closing the feedback loop between the two roles.

8. Authentication and Access Control

- The application supports learner and educator account types.
- Passwords are stored using PHP password_hash() for newly registered accounts.
- Login verification uses password_verify() for hashed passwords.
- The authentication code can upgrade a legacy plaintext password to a secure hash after a successful login.
- PHP sessions store the signed-in user's identity and role.
- Role-aware redirects send educators to the educator experience and learners to the learner experience.
- An authentication guard is included under includes/auth_guard.php for protected access.

9. Database Design

The supplied database dump defines seven principal relational tables. The table design separates users, topics, quizzes, question content, attempts, feedback, and learner recommendations.

Table	Important Fields	Responsibility
user	id, firstName, lastName, emailAddress, password,	Stores learner and educator accounts.

	photoFileName, userType	
topic	id, topicName	Stores quiz subject areas.
quiz	id, educatorID, topicID	Associates a quiz with an educator and topic.
quizquestion	quizID, question, figure, answerA-D, correctAnswer	Stores approved quiz questions.
takenquiz	quizID, score	Stores submitted quiz scores.
quizfeedback	quizID, rating, comments, date	Stores ratings and written feedback.
recommendedquestion	quizID, learnerID, question, answers, correctAnswer, status, comments	Stores learner-proposed questions and educator review.

Implementation note: in the supplied schema, takenquiz and quizfeedback are linked to a quiz but do not include a learnerID field. Therefore, the current database supports aggregate quiz statistics rather than reliable per-learner attempt/feedback history.

10. Data Relationships

- A quiz is associated with one educator and one topic.
- A quiz can contain multiple quiz questions.
- A quiz can have multiple stored attempts and multiple feedback records.
- A recommended question references a quiz and the learner who submitted it.
- When a recommendation is approved, its question data is inserted into quizquestion.

11. Technology Stack

Layer	Technologies Used
Server-side	PHP
Database	MySQL
Database Access	PDO; some project code also uses MySQL-compatible connection patterns
Client-side	HTML5, CSS3, JavaScript
Dynamic Interaction	AJAX / Fetch API, jQuery
UI Libraries	Boxicons, Font Awesome
Local Development	MAMP / MySQL-compatible PHP environment

12. Repository Architecture

The repository is organized by responsibility. The current GitHub structure separates pages, request handlers, API endpoints, shared includes, static assets, uploads, and the database dump.

```
TAMAKAN-main/
├── pages/
│   ├── auth/
│   ├── learner/
│   └── educator/
├── actions/
│   ├── learner/
│   └── educator/
```

```

├── api/
├── includes/
├── assets/
│   ├── css/
│   ├── js/
│   ├── images/
│   └── videos/
├── storage/uploads/
├── database/tamakan.sql
└── README.md

```

13. Main File Responsibilities

Path	Responsibility
pages/auth/	Authentication pages and logout.
pages/learner/	Learner dashboard, quiz, score, and recommendation UI.
pages/educator/	Educator dashboard, question creation/editing, feedback view.
actions/learner/	Processes learner feedback and recommendation submissions.
actions/educator/	Processes educator question CRUD and recommendation decisions.
api/	Returns filtered quiz/educator data for dynamic UI behavior.
includes/	Configuration, access guard, and shared layout components.
assets/	CSS, JavaScript, images, and score-feedback videos.
database/tamakan.sql	Database schema and project data dump.
storage/uploads/	Runtime/user-uploaded files.

14. Validation and Data Handling

- Prepared PDO statements are used in important authentication and database operations.
- Registration requires an educator to select at least one topic.
- Quiz questions use a constrained correct-answer value: A, B, C, or D.
- Recommended questions support an optional uploaded image.
- Recommendation status is constrained to pending, approved, or disapproved.
- Output escaping helpers are used in dashboard views to reduce unsafe HTML rendering.

15. Strengths of the Design

- Two clearly separated user roles with different responsibilities.
- A complete learning loop from quiz delivery to feedback and content contribution.
- Learner-generated content is moderated rather than published directly.
- Approved recommendations become reusable quiz content.
- Topic filtering and asynchronous requests improve navigation.
- Educator dashboards expose useful aggregate indicators such as attempts, average scores, feedback counts, and ratings.

- Motivational result media adds an engagement layer beyond a numeric score.

16. Current Implementation Notes

This documentation describes the behavior supported by the supplied TAMAKAN source code and database. Because the repository was reorganized into functional folders for GitHub presentation, some relative include, asset, form-action, and API paths from the original flat project may require adjustment before this reorganized copy runs directly.

- The current takenquiz table stores quizID and score but not learnerID.
- The current quizfeedback table stores quizID, rating, comments, and date but not learnerID.
- Accordingly, current statistics should be interpreted as quiz-level aggregates unless the schema is extended.
- The repository contains local database configuration under includes/config.php; production deployment should move credentials to environment-specific configuration and keep secrets out of version control.

17. Suggested Future Enhancements

- Add learnerID and timestamps to quiz attempts for personal progress history.
- Associate feedback with the submitting learner when appropriate.
- Add educator analytics over time and per-topic performance trends.
- Introduce stronger upload validation and centralized file handling.
- Add pagination/search for large quiz and recommendation collections.
- Add automated tests for authentication, scoring, CRUD operations, and recommendation approval.
- Move configuration secrets to environment variables.
- Improve responsive and accessibility testing across devices.

18. Conclusion

TAMAKAN demonstrates a collaborative approach to digital assessment. Its central contribution is the controlled feedback cycle between learners and educators: learners practice and respond to existing content, then can actively suggest new questions, while educators retain authority to review and integrate useful contributions. The result is a quiz platform designed not only to assess learning, but also to support continuous improvement of the learning material itself.